

| Question           |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                       | Marks Available |          |          |          |          |          |
|--------------------|-----|------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|                    |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                       | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 7<br>FT<br>2<br>HT | (a) | (i)  |  | Volume = 125<br>Surface area = 294<br>Surface area:volume=3:5/6:10                                                                                                                                                                                                                                                                                                                                                    |                 | 3        |          | 3        | 3        | 3        |
|                    |     | (ii) |  | Diffusion { <u>rate/speed</u> } of the acid into the gelatin cube is the same (1)<br>The bigger the cube the longer decolourisation takes as the acid needs to diffuse {further distance/greater volume}(1) so she is incorrect.<br><br>Or<br><br>Greater/larger surface area:volume ratio in smaller cubes so bigger cubes take longer to decolourise as acid needs to diffuse further for decolourisation to occur. |                 |          | 2        | 2        |          | 2        |
|                    | (b) |      |  | SA:Vol = 3:2 (approx.) / 90:60 / 136:90(1)<br>Diffusion faster (1)<br>Faster gas exchange (1)                                                                                                                                                                                                                                                                                                                         |                 |          | 3        | 3        | 1        |          |
|                    |     |      |  | <b>Question 7 total</b>                                                                                                                                                                                                                                                                                                                                                                                               | <b>0</b>        | <b>3</b> | <b>5</b> | <b>8</b> | <b>4</b> | <b>5</b> |